Project Design Phase
Solution Architecture

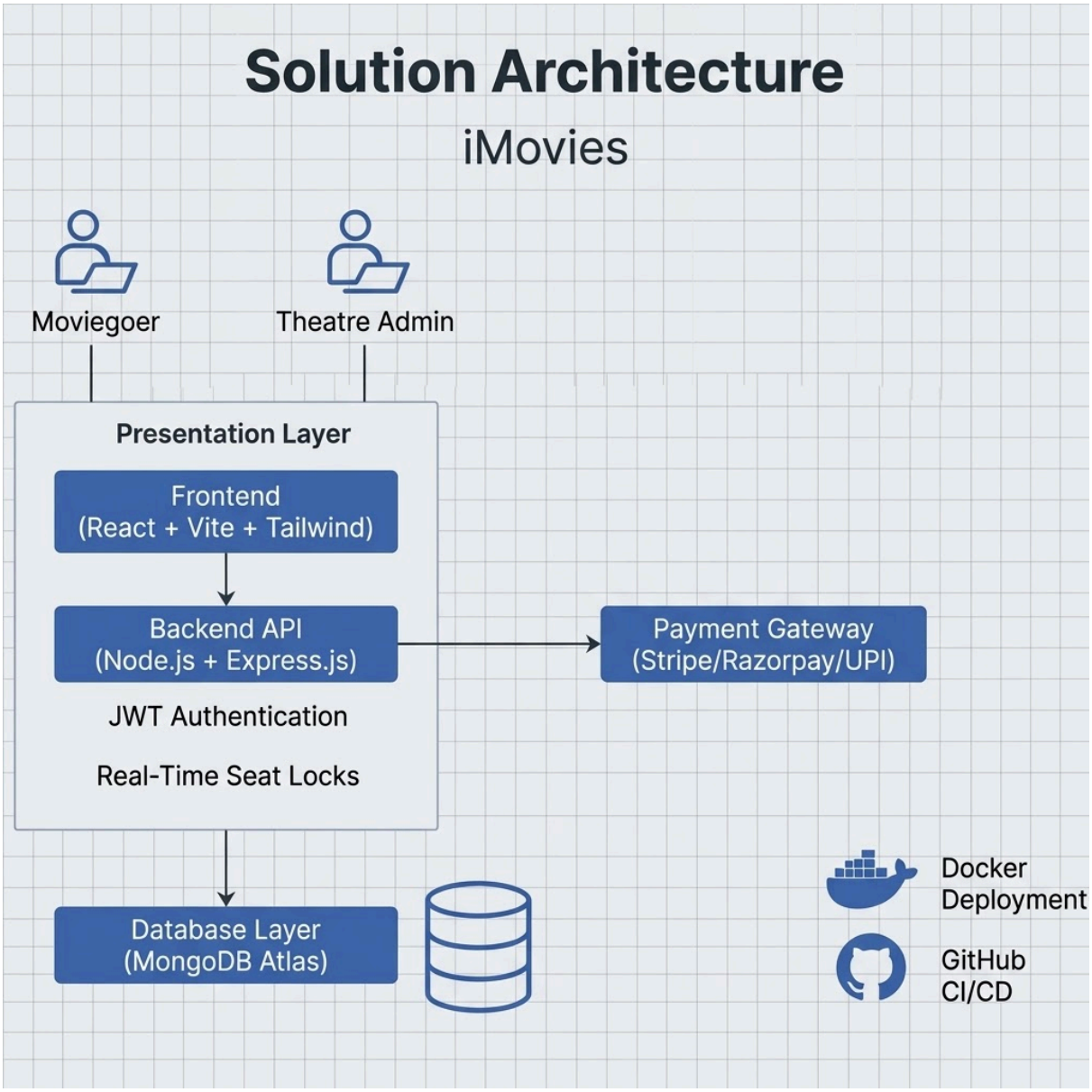
Date	19th October 2026
Team ID	
Project Name	Freelance Finder
Maximum Marks	4 Marks

Solution Architecture:

The solution architecture for iMovies ensures a robust, user-friendly, and scalable platform that connects moviegoers with partner cinema theaters. The architecture focuses on responsive design, secure transactions, atomic seat locking, and automated inventory synchronization to ensure smooth ticketing workflows and high customer satisfaction.

- Seamless movie catalog discovery and showtime scheduling
- End-to-end ticket reservation and booking lifecycle management
- Scalable role-based authentication and authorization (User and Admin tiers)
- Secure and trackable payment processing with digital M-ticket generation
- Reliable real-time seat availability updates and automated notification services

Solution Architecture Diagram:


Solution Architecture
iMovies
Moviegoer
Theatre Admin
Presentation Layer
Frontend
(React + Vite + Tailwind)
Backend API
(Node.js + Express.js)
JWT Authentication
Real-Time Seat Locks
Payment Gateway
(Stripe/Razorpay/UPI)
Database Layer
(MongoDB Atlas)
Docker
Deployment
GitHub
CI/CD